1. Title

Authors

Key Inferences / Points

Critical Review on the Battery State of Charge Estimation Methods for Electric Vehicles (2018)

Rui Xiong et al.

• Reviews lookup-table, coulomb-counting, model-based and data-driven SOC estimation methods.

• Reported errors in reviewed studies: below 5% for an ECM-based approach, below 2% for multi-scale EKF, and below ±1% for an impedance-model-based approach.

• Identifies training-data coverage, model accuracy, cell inconsistency and aging as major challenges.

• Motivates data-driven SOC estimation and lightweight model deployment for our ESP32-S3-based battery monitoring system.

Xiong, R., Cao, J., Yu, Q., He, H., and Sun, F., “Critical Review on the Battery State of Charge Estimation Methods for Electric Vehicles,” IEEE Access, vol. 6, pp. 1832–1843, 2018, doi: 10.1109/ACCESS.2017.2780258.

2. Paper Title

Authors

Key Technical Inferences

Long Short-Term Memory Networks for Accurate State-of-Charge Estimation of Li-ion Batteries

Chemali et al. (2018)

• LSTM-RNN maps voltage, current and temperature directly to SOC, without an explicit battery model or Kalman filter in the estimation pipeline.

• A 500-node LSTM achieved MAE of 0.573% at a temporal depth of 1000 time steps in a fixed-temperature test.

• A single model achieved MAE of 1.606% during a 10–25°C temperature-ramp test.

• Estimation error depends on operating conditions and initialization; the experiments used one NCA cell.

• Motivates data-driven SOC estimation, while embedded deployment and resource requirements must be validated independently.

Chemali et al., “Long Short-Term Memory Networks for Accurate State-of-Charge Estimation of Li-ion Batteries,” IEEE Transactions on Industrial Electronics, vol. 65, no. 8, pp. 6730–6739, Aug. 2018, doi: 10.1109/TIE.2017.2787586.

3. Li-Ion Batteries Parameter Estimation With Tiny Neural Networks Embedded on Intelligent IoT Microcontrollers (2020)

G. Crocioni et al.

• Compared CNN, LSTM, GRU and hybrid models for maximum releasable capacity estimation using 1,462 NASA observations. • CNN-GRU achieved reported maximum RMSE of 0.0488 and MAE of 0.0414. • Evaluated embedded deployment using STM32Cube.AI and benchmarked CNN inference against TFLM. • 8-bit quantization reduced CNN weights and RAM by over 50%, with an accuracy trade-off. • Larger errors on a held-out battery motivate battery-wise generalization testing and ESP32-S3 validation.

Crocioni et al., “Li-Ion Batteries Parameter Estimation With Tiny Neural Networks Embedded on Intelligent IoT Microcontrollers,” IEEE Access, vol. 8, pp. 122135–122146, 2020, doi: 10.1109/ACCESS.2020.3007046.

4. **Paper Title**
   Advances in Batteries, Battery Modeling, Battery Management System, Battery Thermal Management, SOC, SOH, and Charge/Discharge Characteristics in EV Applications (2023)

Authors
R. Ranjith Kumar et al. (2023)

Key Technical Inferences

- Reviews battery models, SOC/SOH estimation techniques and integrated BMS functions.
- **DNN:** Models complex nonlinear battery behaviour but requires greater computational resources; **RNN:** Captures temporal dependencies but may face vanishing/exploding gradient problems.
- Identifies data quality, battery ageing and model complexity as challenges in reliable state estimation.
- Motivates evaluating neural-network accuracy, generalisation and computational requirements for SOC/SOH prediction on the ESP32-S3.

Kumar, R. R., Bharatiraja, C., Udhayakumar, K., Devakirubakaran, S., Sekar, K. S., and Mihet-Popa, L., “Advances in batteries, battery modeling, battery management system, battery thermal management, SOC, SOH, and charge/discharge characteristics in EV applications,” IEEE Access, vol. 11, pp. 105761–105809, 2023, doi: 10.1109/ACCESS.2023.3318121.

5.

Paper Title	Authors	Key Technical Inferences
Comparative Study-Based Data-Driven Models for Lithium-Ion Battery State-of-Charge Estimation	Hussein et al. (2024)	• Compared LR, NN, AE, Transformer, LSTM and RFR using measured $LiFePO_4$ pack data. • RFR achieved the best reported results: MSE = 0.0002765, MAE = 0.0007379, $R^2 = 0.9999996$. • Demonstrated that model complexity alone does not determine SOC estimation performance. • Model size, computational requirements and generalisation require consideration before embedded deployment.

H. M. Hussein, M. Esoofally, A. Donekal, S. M. S. H. Rafin, and O. Mohammed, “Comparative study-based data-driven models for lithium-ion battery state-of-charge estimation,” Batteries, vol. 10, no. 3, Art. no. 89, 2024, doi: 10.3390/batteries10030089.

6. TITLE : A Case Study of a Tiny Machine Learning Application for Battery State-of-Charge Estimation

AUTHOR : Giazitzis et al. (2024)

• ANN and CNN models deployed on a microcontroller using post-training quantization.

• 16-bit quantization reduced model-weight memory by over 50%, with minimal accuracy degradation.

• ANN achieved lower computational cost and memory usage than CNN (20,246 inference cycles; 12.66 kB model memory).

• Battery aging increased SOC estimation errors, highlighting the need for robust models across degradation stages.

• Model selection for ESP32-S3 deployment must consider prediction accuracy, quantization loss, RAM usage and inference cost

S. Giazitzis, M. Sakwa, S. Leva, E. Ogliari, S. Badha, and F. Rosetti, “A case study of a tiny machine learning application for battery state-of-charge estimation,” *Electronics*, vol. 13, no. 10, Art. no. 1964, 2024, doi: 10.3390/electronics13101964.

7. Applying Neural Network to Health Estimation and Lifetime Prediction of Lithium-Ion Batteries

Li et al. (2025)

• Surveys ANN architectures for SOH estimation and RUL prediction from battery-aging data.

• Shallow ANNs: lower computational and memory requirements, but potentially limited capacity to learn complex degradation patterns.

• Deep/hybrid ANNs: greater capacity to learn complex relationships, but higher computational and memory requirements.

• Highlights the need to evaluate prediction error alongside RAM, model size and inference latency for embedded deployment.

Li, P., X. Wu, R. Grosu, J. Hou, M. Ilolov, and S. Xiang, “Applying Neural Network to Health Estimation and Lifetime Prediction of Lithium-Ion Batteries,” IEEE Transactions on Transportation Electrification, vol. 11, no. 1, pp. 4224–4248, 2025, doi: 10.1109/TTE.2024.3457621.

8. Paper Title

Authors

Key Technical Inferences

Embedded Strategy for Battery Module States Estimation Using Tiny Machine Learning Models

Giazitzis et al. (2026)

• Neural-network projection with fine-tuning compresses deep-learning models for TinyML deployment.

• Projected LSTM achieves SOC MAE 0.0114 with a 4.0 kB model.

• Projected CNN-GRU achieves SOH MAE 0.0060 with a 3.4 kB model using ICA features.

• Compression may increase RMSE; GPU-based latency does not establish microcontroller

G. Giazitzis, J. Ferrari, A. J. Woss, S. Badha, F. Rosetti, and E. Ogliari, “Embedded strategy for battery module states estimation using Tiny Machine Learning models,” Journal of Energy Storage, vol. 152, Art. no. 120673, 2026, doi: 10.1016/j.est.2026.120673.